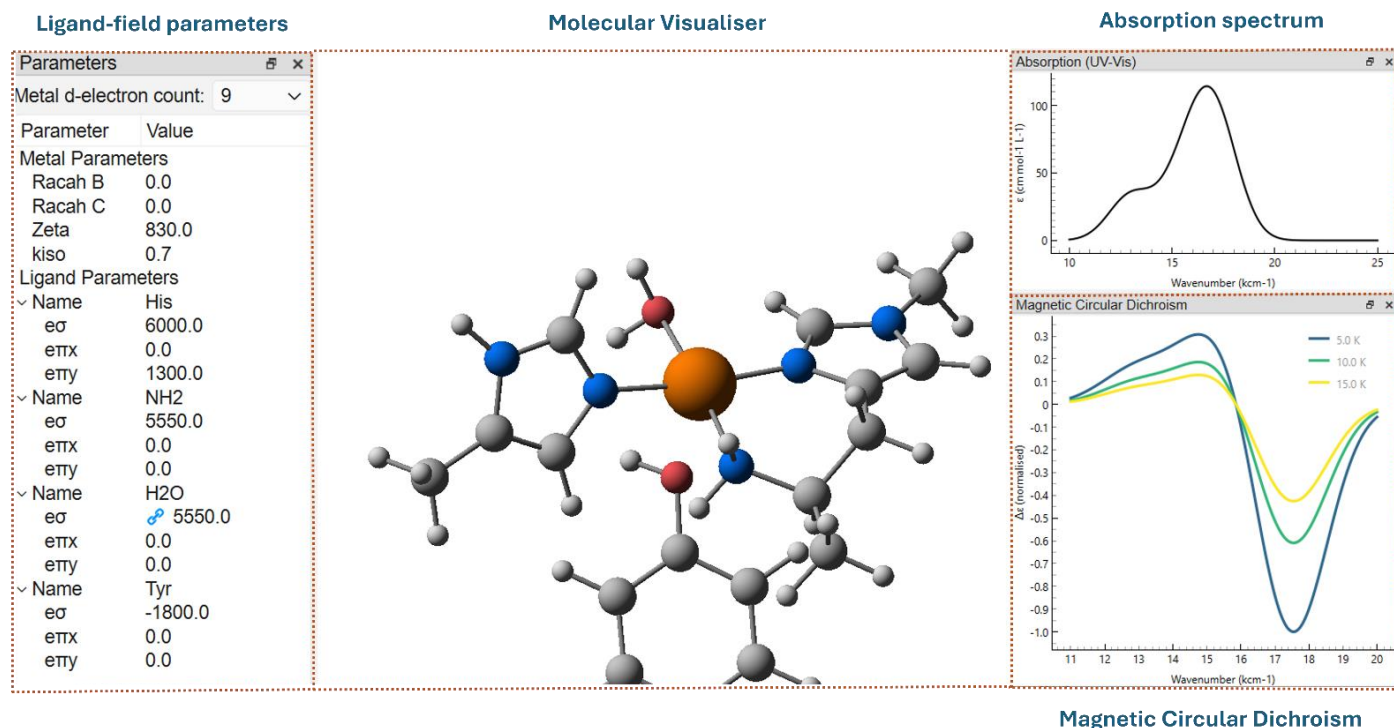


Kestrel 'quickstart' guide



Magnetic Circular Dichroism



Version 1.0

chem-kestrel@york.ac.uk

<https://chem-kestrel.github.io/>

Contributors:

- George Nunn – codebase and GUI development
- Benjamin S. Grace – GUI development and hyperfine module
- Paul H. Walton
- Martin A. Bates

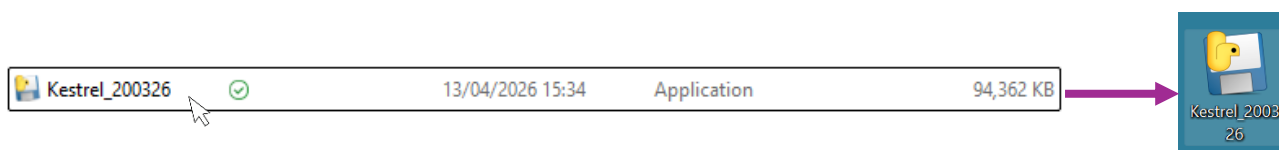
Installation

Before running any calculations in *Kestrel*, the graphical user interface (GUI) or python codebase needs to be installed. This guide will focus on the installation and use of the GUI.

Kestrel will run on computers that support Windows 10 or 11, Linux or MacOS 13 (ventura) and above as a standalone program and requires ~1 Gb of disk space. No further prerequisites are required.

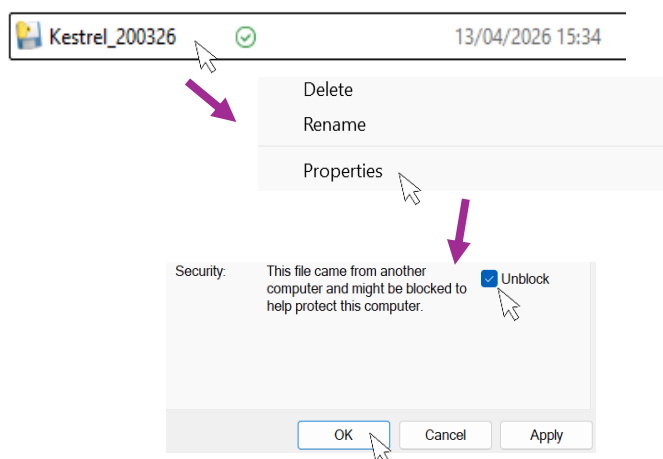
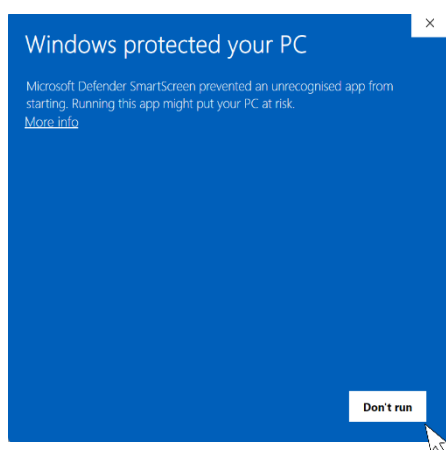
Windows:

- 1) Download the latest version of *Kestrel* from our [website](#). Open 'File Explorer' and navigate to your 'downloads' folder where you will see the *Kestrel* executable.
- 2) Optionally, this file can be dragged onto the desktop for ease of access.
- 3) Double click on the file/thumbnail to launch *Kestrel*.



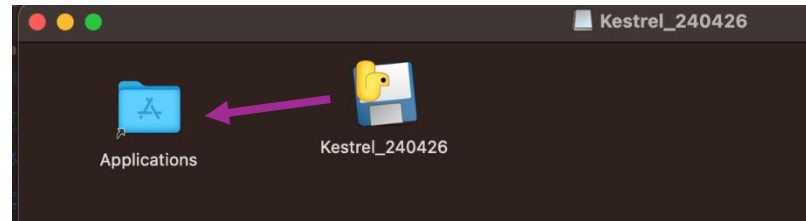
Troubleshooting

- i) Depending on system security settings, the below message may be displayed.
- ii) If this is the case, right click on the .exe file, select 'properties' and check the 'Unblock' box. Click 'OK' and *Kestrel* will run as expected.



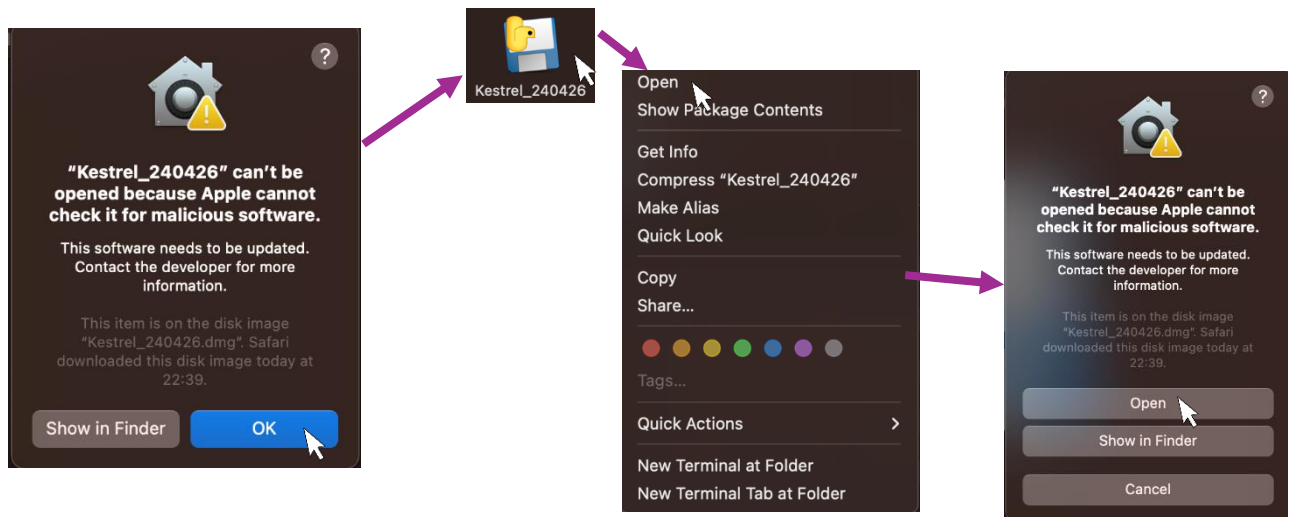
MacOS:

- 1) Download the latest version of *Kestrel* from our [website](#). Open 'Finder' and navigate to your 'downloads' folder where you will see the *Kestrel* .dmg file.
- 2) Double click on the .dmg file and drag the application ("Kestrel.app") to your 'Applications' folder (see below).
- 3) Double click on the thumbnail in the 'Applications' folder to launch *Kestrel*.



Troubleshooting

- i) Depending on system security settings, you may be warned that *Kestrel* cannot be opened.
- ii) If this is the case, click "OK" and then right/double-click on the file/thumbnail. This will open a new menu where you should select 'open'.
- iii) You will then see the same warning as before but this time there will be an option to "Open". Click this and *Kestrel* will launch.



You are now ready to use *Kestrel*. For new users, see pages 6-12 for a 'quick start' guide which walks through a worked example. The remainder of this guide contains detailed documentation on each calculation and option within *Kestrel*.

1.1 Quick Start Guide

In this tutorial, we will import a set of xyz coordinates into *Kestrel*, parameterise the ligand field with a set of e values and display the results of the ligand field calculation.

For this tutorial, we will use the example of $[\text{Cu}(\text{py})_4]^{2+}$ - a square planar Cu^{2+} complex. This relatively simple system has well characterised experimental data allowing *Kestrel*'s results to be validated.

Upon launching *Kestrel*, you will see the screen in figure 1. By navigating to the 'File' menu, previous sessions can be loaded as .kes files or a new molecule created by importing a .xyz file. For this tutorial, we will be importing an .xyz file which can be generated by downloading "*cupy4.xyz*" from the bottom of our website's [documentation page](#).

Alternatively, copy the coordinates on page 13 into a text editor and save with the '.xyz' extension as detailed on page 37

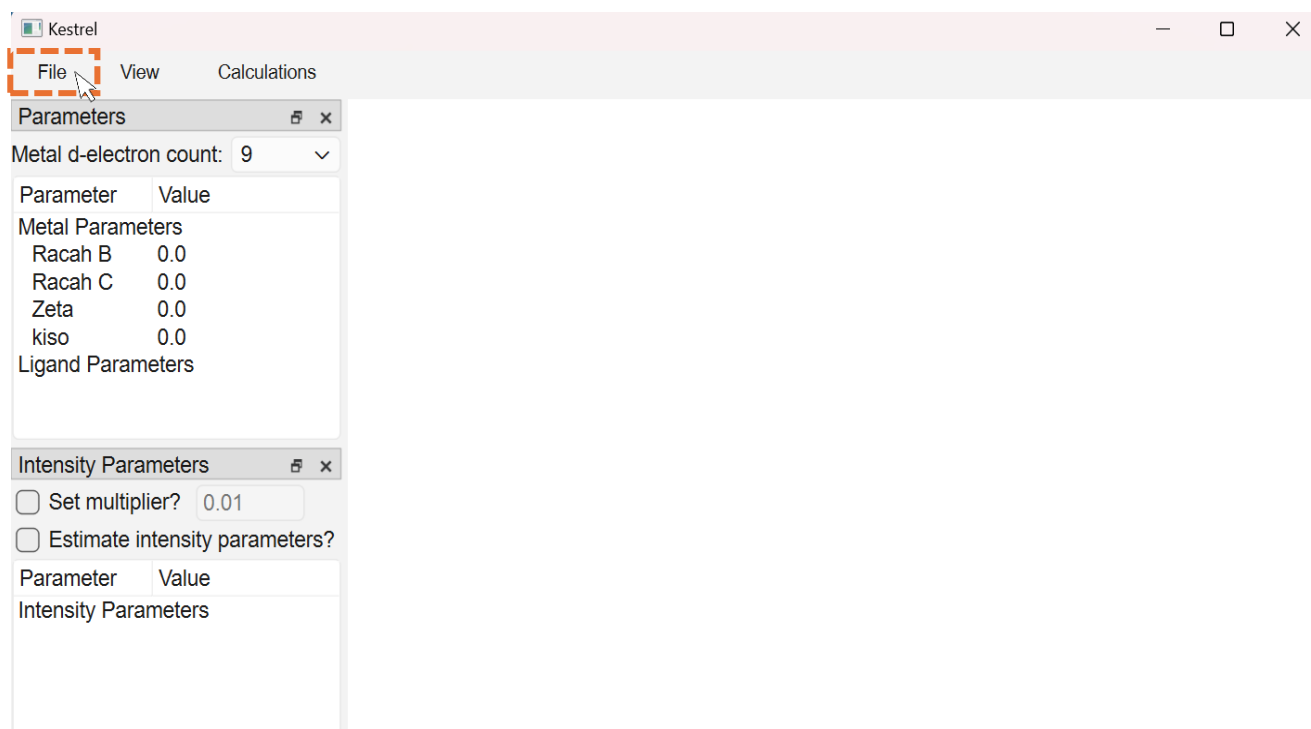


Figure 1: Appearance of Kestrel's GUI upon launching

1.1.1 Loading a set of .xyz coordinates

In *Kestrel*, click on 'File', then 'Import .xyz'. Navigate to a .xyz file, in this case 'cupy4.xyz', and select 'open'. A window similar to that in figure 2 will then appear.

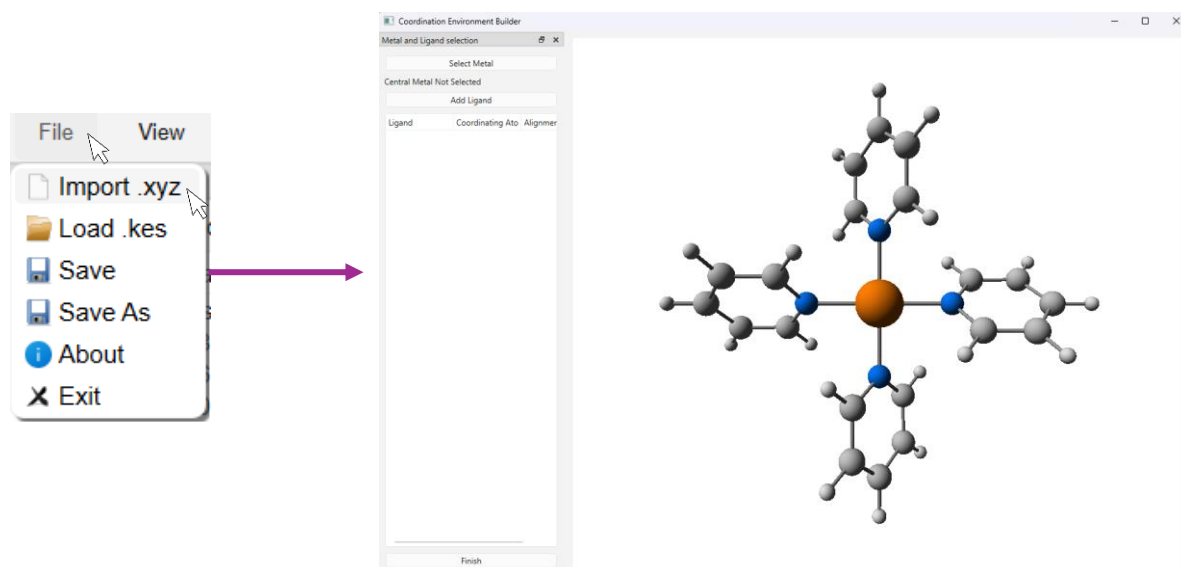
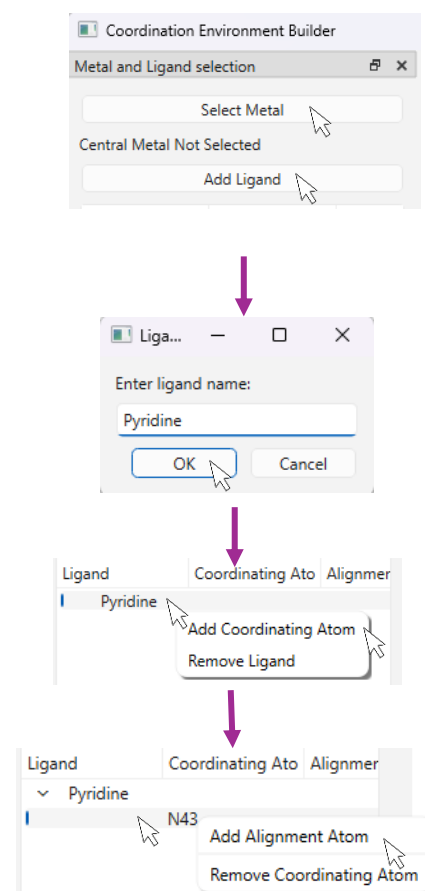


Figure 2: Appearance of 'coordination environment builder' screen

1. First, assign the metal centre by clicking on 'Select Metal' and double-click on the metal atom. The assigned metal will then be shown beneath the 'Select Metal' box.
2. Define each ligand type in the complex by clicking on 'Add Ligand'. Give it a suitable name and select 'OK'. Here we only have one ligand type, pyridine, but multiple ligands can be defined.
3. Assign the coordinating atoms for each ligand type by right-clicking on the ligand's name which will open a menu. Select 'Add Coordinating Atom' and double click on a coordinating atom. For ligands of the same type multiple coordinating atoms can be assigned.
4. For aromatic rings such as pyridine, it is useful to specify the ligand's x/y orientation by right-clicking on a listed coordinating atom which will open a menu. Choose 'Add Alignment Atom' and double-click on the atom the x/y direction is to be aligned with. This ensures the ligand's π system is correctly oriented.



Once finished, you should see the same assignments as in figure 3. If you make a mistake, atoms and ligands can be removed by right-clicking and choosing the 'remove' option from the opened menu. Click 'Finish' once all atoms are assigned to close the window.

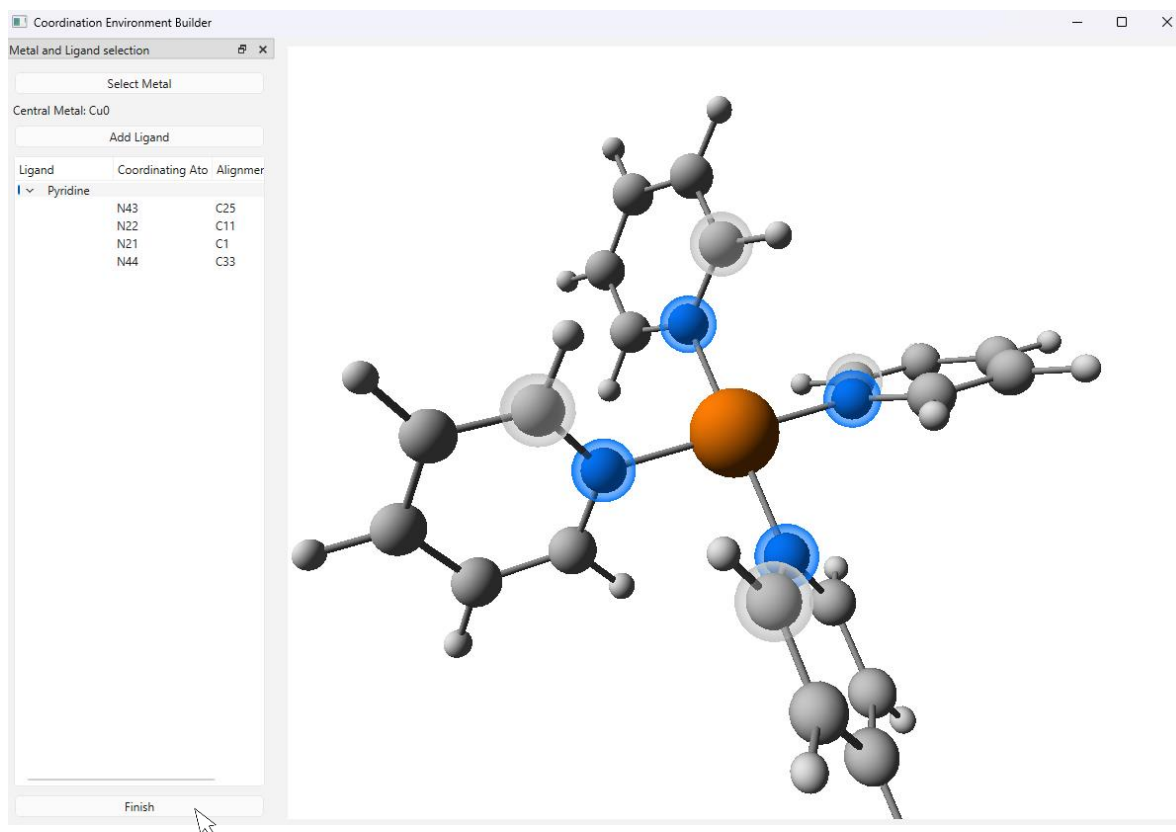


Figure 3: Coordination environment builder with ligand atoms assigned. The coordinating (blue) and orientation atoms (grey) belonging to the pyridine ligand are highlighted.

1.1.2 Assigning Parameter values

The 'Parameters' window on the left of the GUI is where the parameters required for a ligand field calculation are entered. First specify the d-electron count at the top of the panel. The 'Parameters' window is then divided into sections for entering metal and ligand parameters. Parameter values are entered by simply typing a value in each box. The parameter values to enter for this example are shown in figure 4 (dashed box).

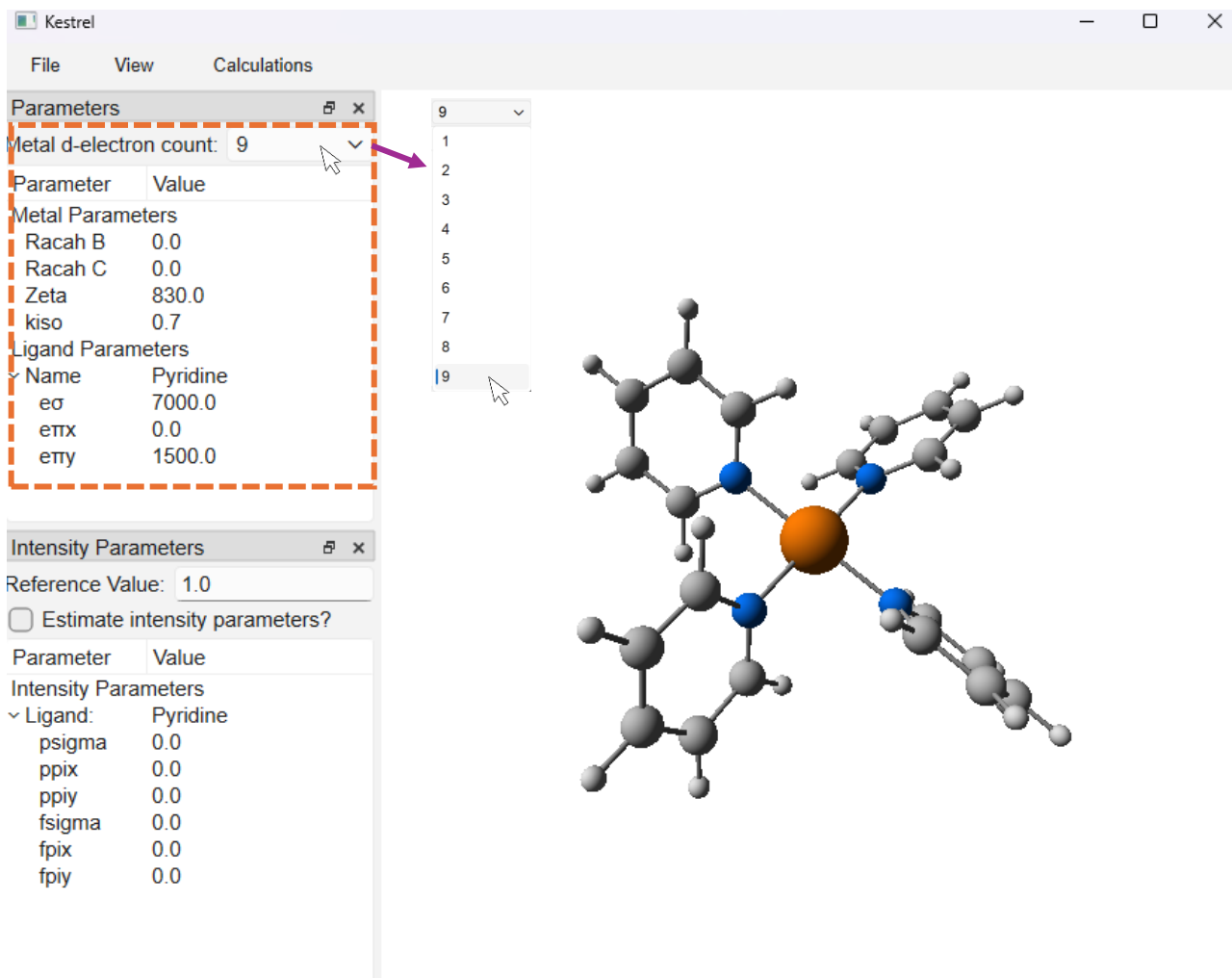


Figure 4: Main screen of Kestrel's GUI after importing .xyz coordinates or loading a .kes file along with the parameter values we will use in this tutorial (dashed box).

1.1.3 Further information: A note on parameter values (optional)

- Metal parameters parameterise the interelectronic repulsion with Racah B and Racah C values. Spin-orbit coupling (SOC) is parameterised with a Zeta value and orbital reduction factor (k_{iso}).
- Ligand parameters describe the strength and type of metal-ligand bond through e parameters. The type of e value (σ or π) describes the bond and its magnitude the strength of bond.¹

For this example the metal centre is Cu^{2+} , which has a d^9 electronic configuration. This can be treated analogously to d^1 , meaning that Racah B and C can be set to zero to represent no interelectronic repulsion since only a single electron (hole) is present. For SOC, we will use the free-ion value for Cu^{2+} of 830 cm^{-1} and an orbital reduction factor of 0.7. For the ligand parameters, we will use an $e\sigma$ value of 7000 cm^{-1} and $e\pi$ value of 1500 cm^{-1} . These values have been determined by fitting to experimental Electron Paramagnetic Resonance (EPR) data.²

Kestrel also has an option to estimate initial ligand parameters by right-clicking on the ligand name and choosing 'Initialise Parameters' (figure 5). The ligand type closest to the one you are parameterising should then be chosen.

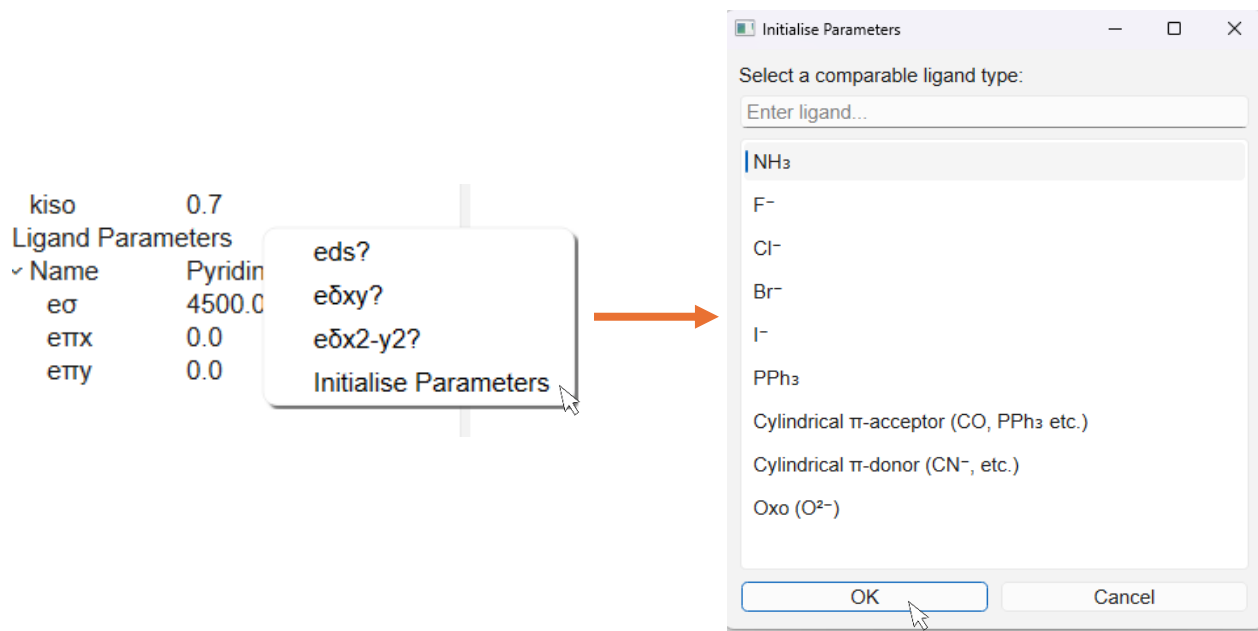


Figure 5: Initialise parameters dialog which is opened after selecting 'Initialise Parameters'. The ligand type that is most comparable to that in the complex should then be chosen to give a set of starting parameters.

1.1.4 Viewing calculation results

To view the results of calculations, click on the ‘Calculations’ option in the top menu where initially, only the Ligand Field Matrix and Eigenfunctions options will be enabled. Clicking on an option will then open a window displaying the results of that calculation.

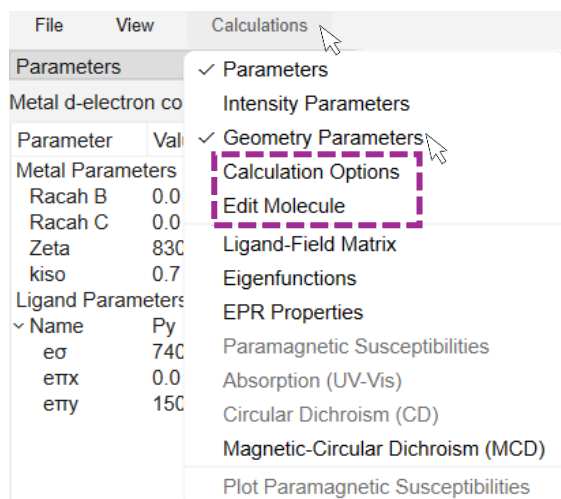


Figure 6: Calculations menu with options to configure a calculation and view its results

Additional calculations can be enabled by clicking on ‘Calculation Options’ which will open a new window. For this example, we will enable EPR calculations by simply checking the ‘Enable EPR Calculations’ and ‘Enable Hyperfine Calculations’ boxes.

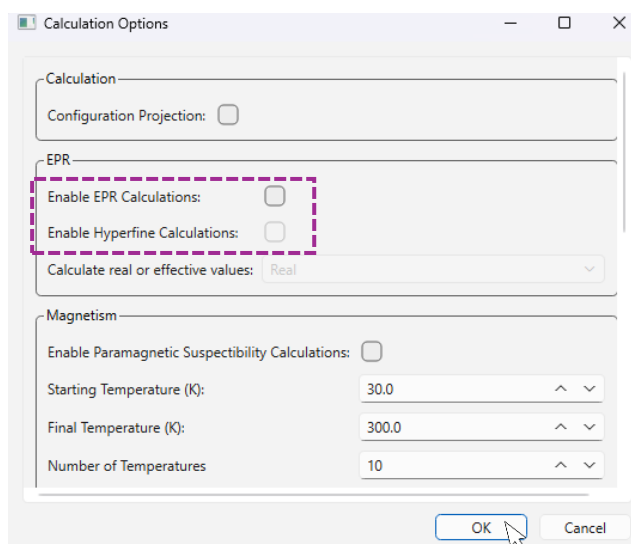


Figure 7: EPR calculation options within the Calculation Options dialog box

Figure 8 shows the results of our calculation (some windows have been moved/resized so that all data are visible):

- 1) EPR g factors and hyperfine values. Experimental g factors are: 2.051, 2.051, 2.262 and hyperfine values are: 48 MHz, 48 MHz, -563 MHz.²
- 2) Ligand field matrix and the energy and composition of each d-orbital. Here we can see the expected square planar splitting of $d_{yz}/d_{xz} < d_{xy} < d_z^2 < d_{x^2-y^2}$ due to the lack of ligands in the z-direction.
- 3) Many-electron eigenfunctions. The energy (and multiplicity) of each electronic state is displayed. The electronic configuration can also be enabled by checking the box 'Configuration Projection'.

Finally, change a parameter value (i.e. increase $\epsilon\sigma$ by 1000 cm⁻¹) – the results will update in real-time.

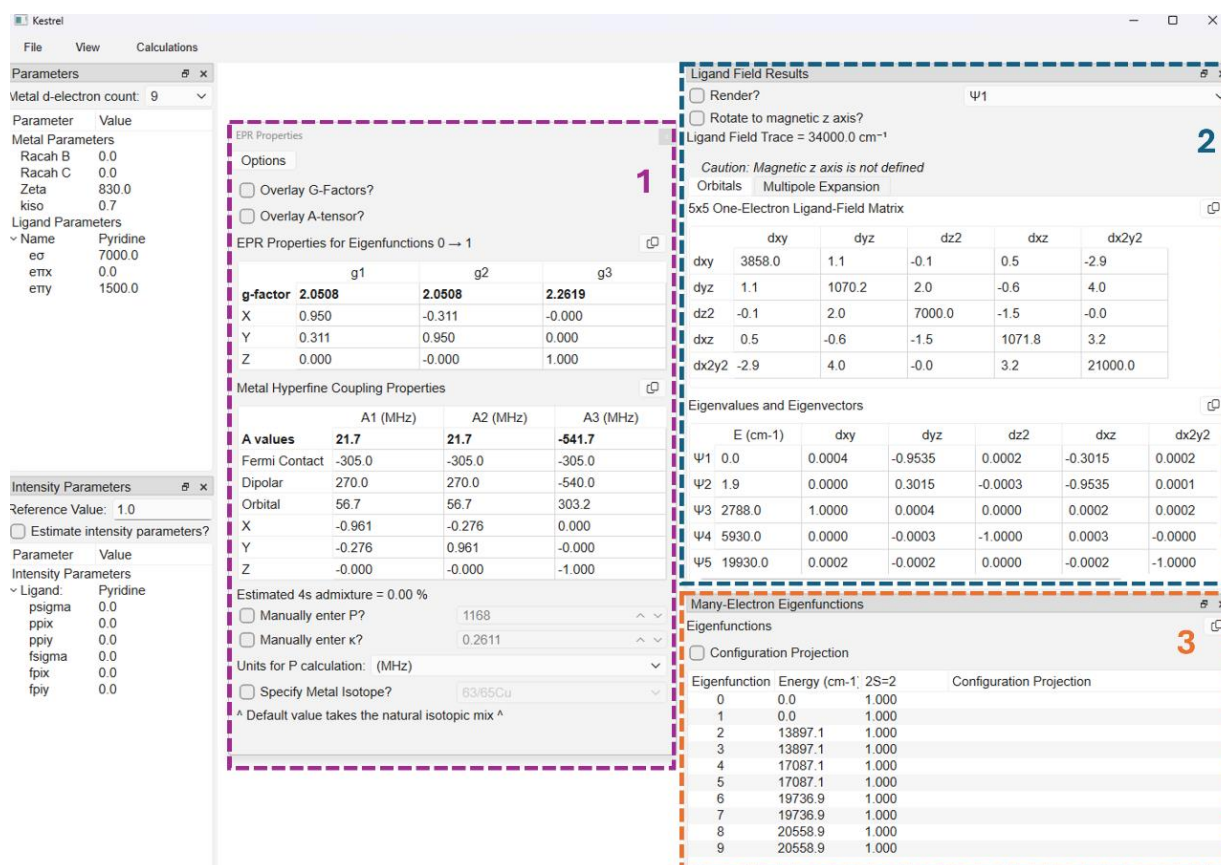


Figure 8: Results of EPR, ligand field and many-electron eigenfunction calculations.

To save the calculation, open 'File' and choose 'Save As'. This then creates a .kes file which stores the current parameter values and molecular geometry.

A .kes file containing the results in figure 8 is available on our [website](#).

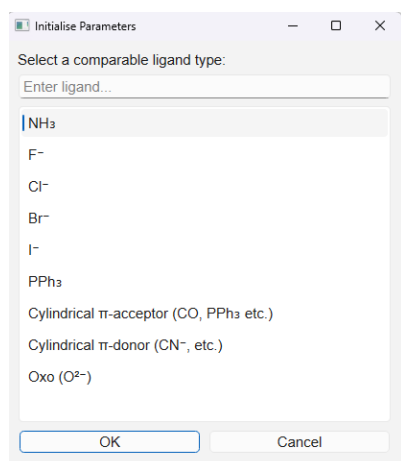
You are now ready to run calculations on any set of xyz coordinates. The remainder of this document contains a detailed description of each feature and calculation but these details are not essential for successful use of Kestrel.

Appendix 1 – How to determine parameter values

Kestrel uses e parameters to describe the bonding interactions between a metal centre and its ligands. Each e parameter describes the type and strength of a bond making each parameter chemically meaningful, with e_σ and e_π describing the strength of σ and π bonding respectively. As an indicative range, e_σ values range from $\sim 3000\text{ cm}^{-1}$ to 10000 cm^{-1} and e_π values from -3000 cm^{-1} to $+3000\text{ cm}^{-1}$ for each metal-ligand bond.

To determine a parameter value, we recommend using either the ‘initialise parameters’ option within *Kestrel* (see page 8) or the table on our [website](#). Choose the ligand type and metal/oxidation state closest to the ligand you are parameterising. For spin-orbit coupling (SOC) use the free-ion value and the average Racah B/C value for the given metal and oxidation state. Values for SOC and Racah B/C can be found on our [website](#).

This provides a starting point and indicates what a chemically feasible set of values would be. The actual values will likely vary by around 10-20% from this starting point.



Search metal or ligand type...							All metals	46 entries
METAL	OXIDATION STATE	LIGAND	$e_\sigma / \text{CM}^{-1}$	$e_{\pi x} / \text{CM}^{-1}$	$e_{\pi y} / \text{CM}^{-1}$	REFERENCE		
Co	+2	F ⁻	5,300	2,200	2,200	Buchhorn et al. (2022)		
Co	+2	Cl ⁻	3,600	900	900	Bencini et al. (1984)		
Co	+2	Cl ⁻	3,150	1,100	1,100	Buchhorn et al. (2022)		
Co	+2	Br ⁻	3,100	775	775	Bencini et al. (1984)		
Co	+2	I ⁻	2,700	900	900	Buchhorn et al. (2022)		

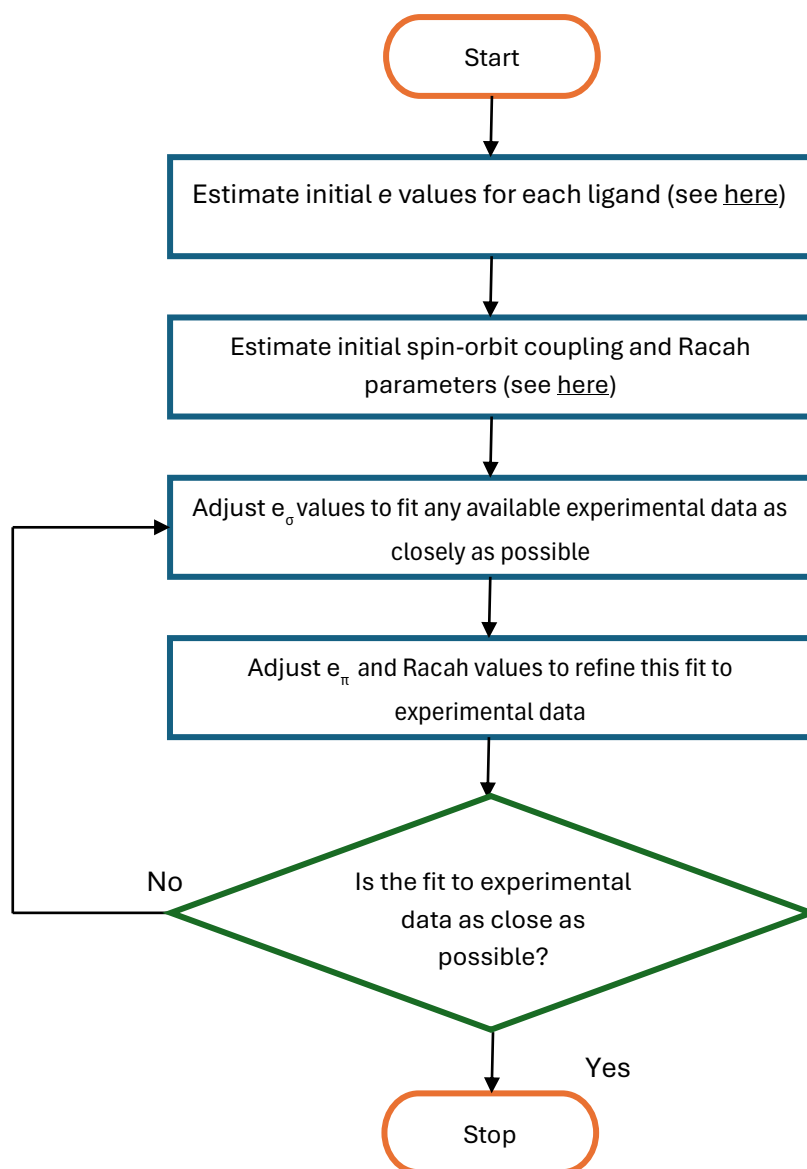
↓ Show all entries

↑ Values are taken from the cited publications. If you have parameter values from a published study that are not listed here, please [contact us](#) to have them included.

Initialise parameters box in Kestrel (left) and parameter table on our website (right) from which initial parameter values can be obtained.

The initial e values should then be adjusted to fit any available experimental data. We recommend that you adjust e_σ values first to get an approximate fit to experimental data and then fine tune by adjusting e_π parameters along with any interelectronic repulsion parameters. This generally requires an iterative process to obtain a final set of parameter values which give the closest fit to experimental data.

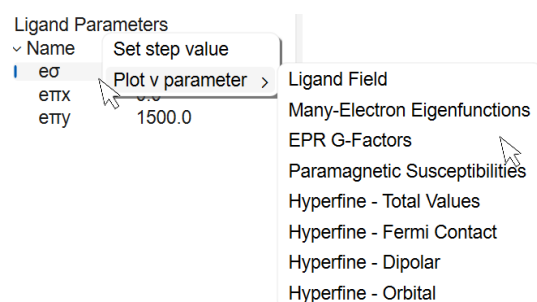
On the next page we have set out a flow chart to illustrate this process of determining parameter values in *Kestrel*.



What if no or limited experimental data are available?

Firstly, we would recommend fitting to whatever experimental data are available, even if limited it will still help to refine your parameter values. If there is no experimental data, then we recommend using estimated values for the ligand and metal combination closest to that which you are parameterising to provide a general estimate for each calculation result.

In this situation, it is often useful to plot variation diagrams by right-clicking on a parameter value and selecting 'plot v parameter'. Since it is expected that a starting parameter would vary by up to 20%, this shows how such a variation in parameter value would affect the calculation results and indicates the possible error.



[CuPy4]2+ .xyz coordinates

Cu 0.00000000 0.00000000 0.00000000
C -0.695174739 2.703797661 -0.935667100
H -1.252849847 2.112386426 -1.661786403
C 0.696234162 2.704292804 0.934554740
H 1.252748139 2.113413780 1.661978708
C 0.710592471 4.094287803 0.969873234
H 1.274833194 4.606919765 1.748082093
C 0.002238170 4.803410619 -0.002494345
H 0.003157740 5.893671184 -0.003448466
C -0.707342372 4.093765099 -0.973548961
H -1.270865502 4.605843838 -1.752638327
C -2.705163278 0.695714907 0.934039059
H -2.114604722 1.251200921 1.662510065
C -4.095195580 0.710639422 0.968342019
H -4.608161030 1.274604700 1.746531287
C -4.803845710 0.003299507 -0.005089467
H -5.894103231 0.004604114 -0.006879539
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H -2.112077049 -1.251167681 -1.663681185
N 0.000000000 2.020299387 0.000000000
N -2.020685339 0.000061870 -0.000598595
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H 2.112131948 -1.255415688 1.660154620
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H 2.113632342 1.254455924 -1.660384628
C 4.094407688 0.711018657 -0.968991246
H 4.607146773 1.276153567 -1.746479720
C 4.803395673 0.001277728 0.002463020
H 5.893656990 0.001972041 0.003404639
C 4.093617228 -0.709406254 0.972614170
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H -1.271697998 -4.605128671 1.754604561
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H -1.253849671 -2.111733437 1.662074837
N 2.020290525 -0.000386831 0.000000000
N -0.000206020 -2.020677521 0.000793206

References

1. M. Gerloch, *Magnetism and ligand-field analysis*, Cambridge University Press, Cambridge, 1st Edition edn., 1983.
2. F. Neese, *The Journal of Chemical Physics*, 2003, **118**, 3939-3948.

Citation Information

If you use Kestrel in your work, we kindly ask that you provide the following citation:

- G. Nunn, B. S. Grace, M. A. Bates and P. H. Walton, Kestrel [Computer software]. <https://github.com/chem-kestrel/Kestrelpy>

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